A Robust Hybrid Framework for Industrial Pallet Defect Detection using Evolutionary Deep Learning

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ABSTRACT

Industrial pallet inspection is a critical quality control step in manufacturing and logistics, yet manual inspection remains error-prone and inconsistent at scale. This poster presents a hybrid automated defect detection system built on Evolutionary Deep Learning combining evolutionary hyperparameter optimization with deep learning classification [6][7] and hand-crafted OpenCV segmentation. ES(1+1) and Genetic Algorithm optimization are applied to VGG16 [1] fine-tuning hyperparameters, achieving 99.5% test accuracy and 98% deployment accuracy outperforming all fixed-hyperparameter baselines. We evaluate eight model configurations on 7,486 images under raw and segmented input conditions and introduce a deployment accuracy metric computed on 100 live unseen webcam images. Results show that evolutionary optimization closes the test-to-deployment gap more effectively than architecture choice alone, and that HSV segmentation with perspective warp normalization is the critical preprocessing step that enables evolutionary optimization to generalize across camera conditions.

INTRODUCTION & Research Goals

Problem: Manual pallet inspection is slow and error-prone. Deep learning classifiers [6][7] trained in lab conditions frequently fail in real deployment due to domain shift [7] changes in camera, lighting, and background between training and live environments. **The Hidden Risk:** A model scoring 99% on a held-out test set can drop to 50% accuracy on a live camera identical to random guessing. Standard test accuracy does not reveal this failure. **Deployment Accuracy:** We introduce a metric computed on 100 live webcam images collected under unseen conditions. This measures true industrial readiness not lab performance. **Our Approach:** HSV segmentation removes background variability before classification. We compare 8 model configurations and apply evolutionary optimization to close the test-to-deployment gap.

Fig. 1. live camera setup for deployment test

RESEARCH GOALS

- Build an HSV segmentation pipeline to remove background variability and improve cross-camera generalization
- Compare raw vs segmented input across 8 model configurations to quantify segmentation's effect on test and deployment accuracy
- Apply ES(1+1) [3], and Genetic Algorithm [4] to optimize hyperparameters beyond fixed baselines
- Introduce a deployment accuracy metric computed on 100 live unseen webcam images distinct from test set accuracy
- Identify which factor segmentation, architecture, or EC optimization closes the test-to-deployment gap most

DATASET

7,486 images · 2 classes: Good (all parts present) / Bad (missing or misoriented parts) 70% train (5,046) · 15% validation (1,196) · 15% test (1,244) · Balanced 50/50 Sources: lab camera, iPhone, Android, laptop webcam Augmentation: brightness, contrast, saturation, hue, ±10° rotation, no flips [6]

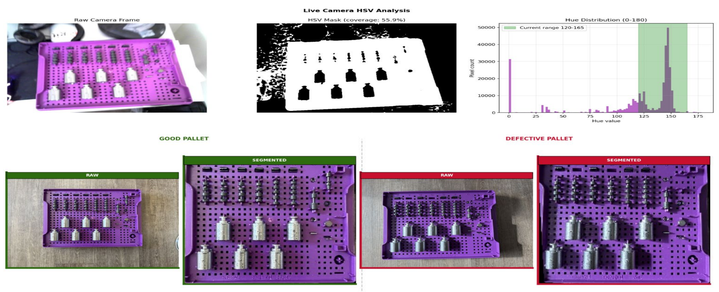

Fig2, 3. HSV Segmentation & examples of raw and segmented image

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METHODS

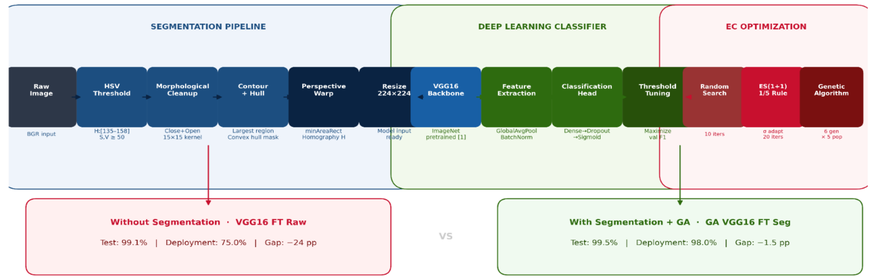

Fig. 3. System pipeline HSV segmentation, deep learning classification, and evolutionary optimization.

SEGMENTATION PIPELINE

HSV Thresholding: Isolate purple pallet using hue range [135–158], S,V ≥ 50. HSV chosen over RGB because hue is stable under lighting changes. mask(x,y) = 1 if (H_low) ≤ (H) ≤ (H_up) ∧ (S_low) ≤ (S) ∧ (V_low) ≤ (V). **Morphological Cleanup**: Closing fills holes inside pallet. Opening removes background noise. 15×15 rectangular kernel. **Contour Detection**: Largest contiguous region selected as pallet. **Convex Hull Mask**: Hull filled as solid polygon covering entire pallet including components. Background set to pure black. **Perspective Warp**: minAreaRect captures orientation angle θ. Homography H maps rotated corners to upright rectangle. H = getPerspectiveTransform(src, dst). **Resize**: Output resized to 224×224.

MODEL ARCHITECTURES

Transfer Learning Phases: Feature Extraction (FE): backbone frozen, only classification head trained. **Fine-Tuning (FT):** top backbone layers unfrozen and retrained at low learning rate (1e-5) to prevent catastrophic forgetting. **Baseline CNN**: Custom 3-block network. Each block: Conv2D → BatchNorm → ReLU → MaxPool. Head: GlobalAvgPool → Dense(256) → Dropout → Dense(1, sigmoid).

EfficientNetB0 FE [2]: Frozen ImageNet backbone. Head: GlobalAvgPool → BatchNorm → Dense(256) → Dropout(0.4) → Dense(128) → Dropout(0.3) → Sigmoid. LR: 1e-3. **EfficientNetB0 FT [2]**: Same as FE with top layers unfrozen. LR: 1e-5. **VGG16 FE [1]**: Frozen VGG16 backbone. Same head as EfficientNetB0 FE. **VGG16 FT [1]**: Block5 (last 4 layers) unfrozen. Two-stage: FE first then FT. LR: 1e-5. **Training settings (all models):** Adam · Binary cross-entropy · Max 60 epochs · Early stopping patience 10 · ReduceLROnPlateau factor 0.5 · Threshold tuned on val set to maximize macro F1

ES(1+1) [3] 20 iterations Single parent → mutate → evaluate → accept if fitness ≥ parent − 0.02 (tolerance accounts for stochastic variation in validation F1 due to random weight initialization) $x' = x + \sigma \cdot N(0,1)$ σ adapts via 1/5 rule: $\sigma \times 1.22$ if accept rate > 1/5 · $\sigma \times 0.82$ if < 1/5

GA [4] 5 individuals 6 generations Tournament selection (k=3) Uniform crossover · Mutation rate: 0.3 · Elitism: best individual carried forward Fitness function: f(θ) = macro F1 score on the validation set, where θ represents the hyperparameter configuration being evaluated

Hyperparameters optimized by ES(1+1) [3] and GA [4] on VGG16 fine-tuning: Learning rate · Unfrozen layers · Dense units · Dropout rate · Batch size

Hyperparameter	*Range*	*Type*
Learning rate	[1e-6, 1e-4]	Continuous (log)
Batch size	{16, 32}	Discrete
Unfrozen layers	{4, 8, 12, 16}	Discrete
Dense units	{128, 256, 512}	Discrete
Dropout rate	[0.1, 0.5]	Continuous

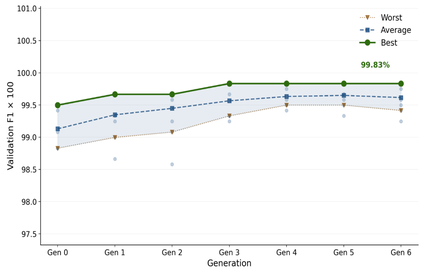

Fig 4. GA fitness evolution

DEPLOYMENT ACCURACY

Deployment Accuracy = correct live predictions / total live images × 100%

How measured: 50 good + 50 bad pallet images captured live from a laptop webcam different camera, lighting, angle, and background from training. Never seen during training. Each image run through all models independently.**Why it matters:** A model scoring 99% test accuracy but 50% deployment accuracy is useless in production 50% on a balanced dataset equals random guessing. Gaps up to 49pp (percentage points) were observed. For any industrial vision system, deployment accuracy is the only honest measure of readiness.

RESULTS

Key Numbers

- 99.5% Best test set accuracy (ES VGG16 Seg and GA VGG16 Seg)
- 98.0% Best deployment accuracy (VGG16 FT Seg and GA VGG16 Seg)
- 50.0% Worst deployment accuracy (Baseline CNN Raw, ES Raw) same as random guessing
- 7,486 Total training images
- +48pp (percentage points) Maximum improvement in deployment accuracy from raw to segmented (Baseline: 50% → 93%)

Model	Test Accuracy	Deployment Accuracy
Baseline CNN Raw	90.3%	50.0%
Baseline CNN Seg	99.4%	93.0%
EffNetB0 FE Raw	94.3%	77.0%
EffNetB0 FE Seg	98.8%	88.0%
VGG16 FT Raw	99.1%	87.0%
VGG16 FT Seg	98.4%	98.0%
ES VGG16 Seg	99.5%	96.0%
GA VGG16 Seg	99.5%	98.0%

Fig. 6. Results Table with Test & Deployment Accuracies

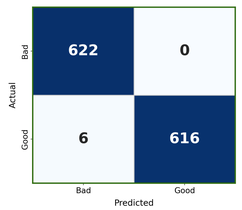

Fig. 5. Confusion Matrix GA VGG16 Seg

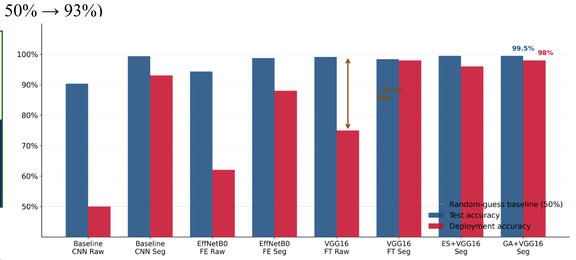

Fig. 7. Bar Chart comparing raw and segmented image

DISCUSSION

Evolutionary optimization drives the best results. GA-optimized VGG16 [1][4] (98% deployment) and ES-optimized VGG16 [1][3] (96% deployment) outperform all fixed-hyperparameter baselines, confirming EC optimization meaningfully improves generalization beyond architecture alone.

Models do not generalize across camera conditions. VGG16 FT Raw [1] scores 99.1% test but only 75% live a 24 pp (percentage points) gap. Baseline CNN Raw scores 90.3% test but 50% live, random guessing. High test accuracy is necessary but not sufficient for industrial readiness.

Segmentation is the critical enabling factor. Segmented models averaged 88.3% deployment vs 67.0% raw a 21 pp gap improved in every single model pairing.

Limitations. 100 images from one webcam. Pipeline fails if pallet color changes or lighting is severely degraded.

CONCLUSION

ES(1+1) [3] and GA [4] optimization is the primary contribution, GA-optimized VGG16 [1][4] achieves 99.5% test and 98% deployment accuracy, outperforming all fixed-hyperparameter models. HSV segmentation is the critical enabling factor, segmented models averaged 88.3% vs 67.0% raw, a 21 pp (percentage points) improvement. All three combined, segmentation + pretrained VGG16 [1] + evolutionary optimization, produce the best result: 99.5% test, 98% deployment, 1.5 pp gap. Deployment accuracy on live unseen images is the correct measure of industrial readiness, test accuracy alone is not sufficient.

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